Seminar 1 - Analytic Geometry

ECE 447: Data Analysis and Machine Learning for Engineers

Bahareh Tolooshams, Ph.D.
Assistant Professor,
University of Alberta
btolooshams@ualberta.ca
https://btolooshams.github.io

January 07, 2026

Land Acknowledgment

The University of Alberta acknowledges that we are located on Treaty 6 territory, and respects the histories, languages, and cultures of First Nations, Métis, Inuit, and all First Peoples of Canada, whose presence continues to enrich our vibrant community.

This Seminar Covers ...

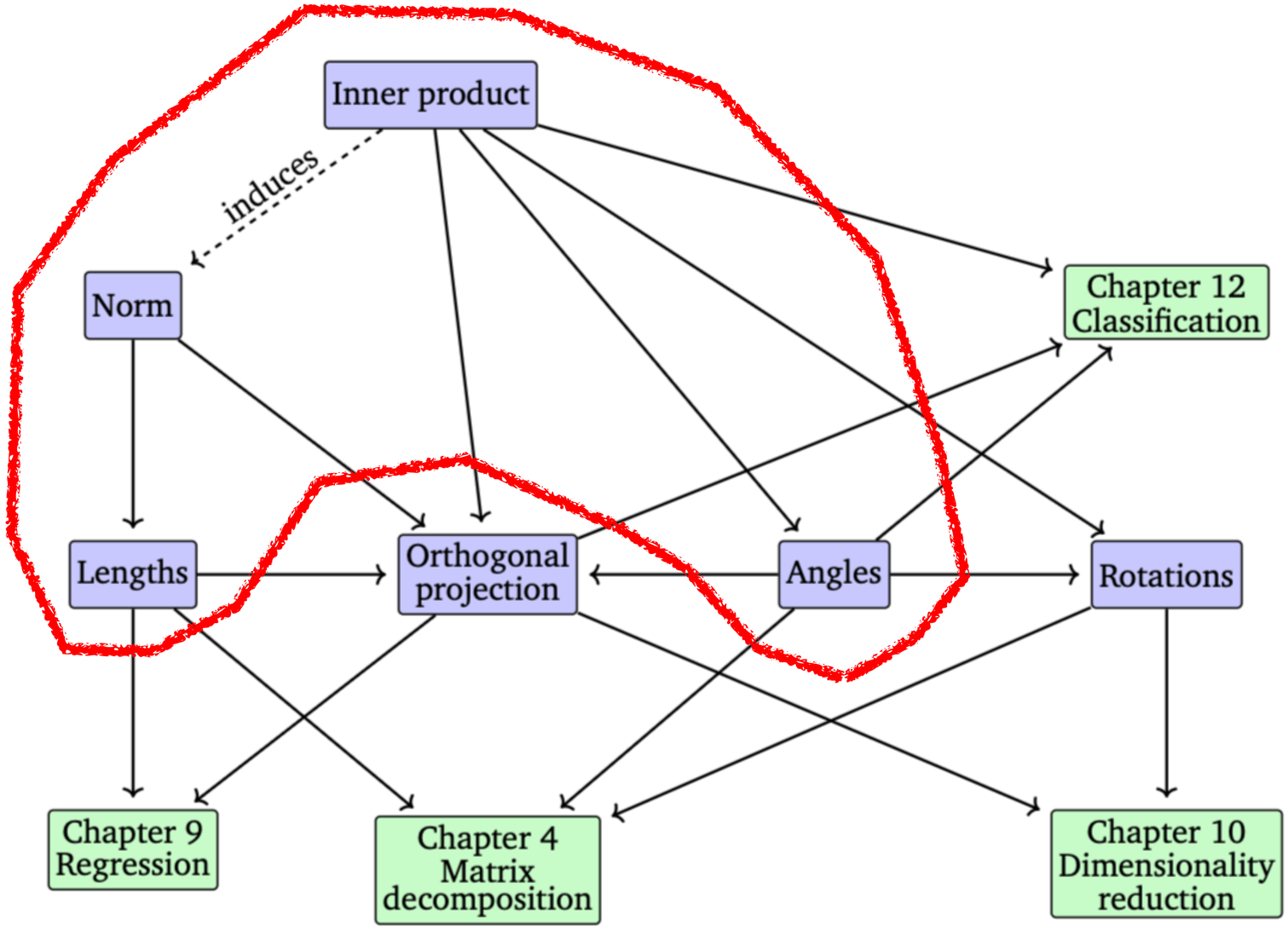


figure from Machine Learning A Bayesian and Optimization Perspective textbook.

We Deal with **Optimization** for Machine Learning Methods

Ridge Regression

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \quad \|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2^2 \quad + \quad \lambda\|\boldsymbol{x}\|_2^2$$

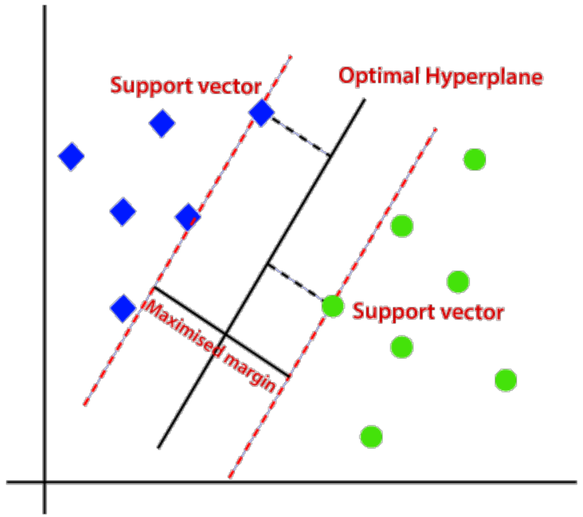
Support Vector Machine (SVM)

$$\min_{\boldsymbol{\theta}_1,\theta_0,\zeta} \quad \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 + C\sum_{i=1}^{n}\zeta_i$$

$$\text{subject to} \quad \zeta_i \geq 0, \quad i=1,\ldots,n$$

$$y_i(\boldsymbol{x}_i^\top\boldsymbol{\theta}_1+\theta_0) \geq 1-\zeta_i, \quad i=1,\ldots,n$$

where $y_i \in \{-1,1\}$, and $\boldsymbol{x}_i \in \mathbb{R}^p$.



https://medium.com/@ambika199820/support-vector-machine-svm-algorithm-explained-c8facc10bcfa.

Metric Space

A **metric** d on a set X is a function $d : X \times X \to [0,\infty)$ for that for all $x, y, z \in X$:

(I) $d(x, y) \geq 0$ *(non-negative)*

(II) $d(x, y) = 0$ if and only if $x = y$

(III) $d(x, y) = d(y, x)$ *(symmetry)*

(IV) $d(x, z) \leq d(x, y) + d(y, z)$ *(triangle inequality)*

Intuition. Function to measure the distance between two data samples.

Euclidean Distance Metric

Let $d : \mathbb{R}^2 \times \mathbb{R}^2 \to [0,\infty)$:

$$d(\boldsymbol{x},\boldsymbol{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

where $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$.

The Euclidean metric $d : \mathbb{R}^n \times \mathbb{R}^n \to [0,\infty)$ on $\mathbb{R}^n$ is defined by

$$d(\boldsymbol{x},\boldsymbol{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

Absolute Value and Manhattan Distance Metric

The **absolute value** distance metric $d : \mathbb{R} \times \mathbb{R} \to [0,\infty)$ on $\mathbb{R}$ is defined by

$$d(x, y) = |x - y|$$

The **Manhattan** metric $d : \mathbb{R}^2 \times \mathbb{R}^2 \to [0,\infty)$ on $\mathbb{R}^2$ is defined by

$$d(\boldsymbol{x}, \boldsymbol{y}) = |x_1 - y_1| + |x_2 - y_2| \qquad \textit{(also know as } \ell_1\textit{)}$$

where $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$.

Minkowski Distance Metric

The **Minkowski** distance metric $d : \mathbb{R}^n \times \mathbb{R}^n \to [0,\infty)$ on $\mathbb{R}^n$ is defined by

$$d(x, y) = \left(\sum_{k=1}^{n} |x_k - y_k|^p \right)^{\frac{1}{p}}$$

where $\boldsymbol{x} = [x_1, x_2, \ldots, x_n]^\top$ and $\boldsymbol{y} = [y_1, y_2, \ldots, y_n]^\top$.

$p = 1 \to \ell_1$

$p = 2 \to \ell_2$ or Euclidean

Cosine Distance Metric

The **Cosine** distance metric $d : \mathbb{R}^n \times \mathbb{R}^n \to [0,\infty)$ on $\mathbb{R}^n$ is defined by

$$\cos(\boldsymbol{x}, \boldsymbol{y}) = \frac{\boldsymbol{x}^\top \boldsymbol{y}}{\|\boldsymbol{x}\| \|\boldsymbol{y}\|} = \frac{\sum_{k=1}^n x_k y_k}{\sqrt{\sum_{k=1}^n x_k^2} \sqrt{\sum_{k=1}^n y_k^2}}$$

where $\boldsymbol{x} = [x_1, x_2, \ldots, x_n]^\top$ and $\boldsymbol{y} = [y_1, y_2, \ldots, y_n]^\top$.

Intuition: Function to measure the angle between two vectors.

Inner Product

Dot product is an inner product defined by

$$\boldsymbol{x}^\top \boldsymbol{y} = \sum_{k=1}^{n} x_k y_k$$

where $\boldsymbol{x} = [x_1, x_2, \ldots, x_n]^\top$ and $\boldsymbol{y} = [y_1, y_2, \ldots, y_n]^\top$.

In this course, we use to notation of inner product $\langle \cdot , \cdot \rangle$ interchangeably for dot product.

Norm

A **norm** on a real vector space V is a function $\|\cdot\| : V \to \mathbb{R}$ satisfying

(I) $\|\boldsymbol{u}\| > 0$ for any non-zero vector $\boldsymbol{u} \in V$ *(positive)*

(II) $\|\boldsymbol{u}\| = 0$ if and only if $\boldsymbol{u} = 0$

(III) $\|a\boldsymbol{u}\| = |a|\|\boldsymbol{u}\|$ for any $a \in \mathbb{R}$ and $\boldsymbol{u} \in V$ *(homogeneous)*

(IV) $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$ for any $u, v \in V$ *(triangle inequality)*

*Norms are **convex**, hence, useful in optimization.*

We will learn about convexity soon!

General ℓ_p-Norms

ℓ_p-norm

$$\|\boldsymbol{x}\|_p = \left(\sum_{k=1}^{n} |x_k|^p \right)^{\frac{1}{p}}$$

for some $p \geq 1$ and $\boldsymbol{x} = [x_1, x_2, \ldots, x_n]^\top$.

Special cases:

ℓ_1-norm

$$\|\boldsymbol{x}\|_1 = \sum_{k=1}^{n} |x_k|$$

ℓ_2-norm

$$\|\boldsymbol{x}\|_2 = \sqrt{\sum_{k=1}^{n} x_k^2}$$

or

$$\|\boldsymbol{x}\|$$

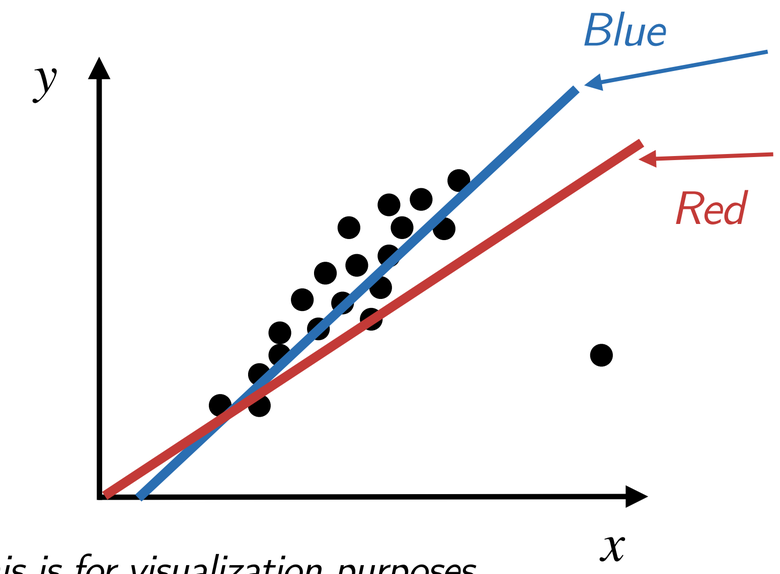
ℓ_∞-norm

$$\|\boldsymbol{x}\|_\infty = \max_k \{ |x_k| \}$$

Imaging you are doing linear regression (fitting a hyperplane to your data), which tries to minimize the distance of points in the y-axis to your line. Based on your understanding of ℓ_1 and ℓ_2 which line represents which best?

Intuition on Norms

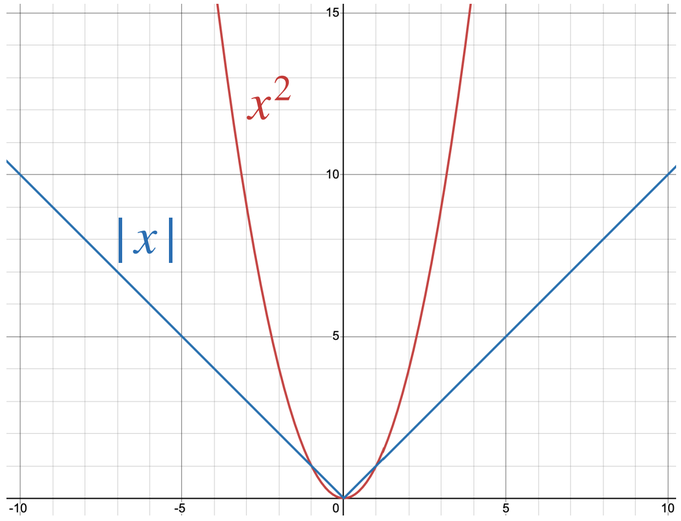
Which norms (ℓ_1, ℓ_2, ℓ_∞, ...) should I minimize as distance metric in ML?



This is for visualization purposes.

ℓ_1-norm based distance $\sum_{i=1}^{n} |y_i - f(x_i)|$ for linear regression

ℓ_2-norm based distance $\sum_{i=1}^{n} (y_i - f(x_i))^2$ for linear regression



https://www.desmos.com/calculator

ℓ_2 penalizes large errors more than ℓ_1.

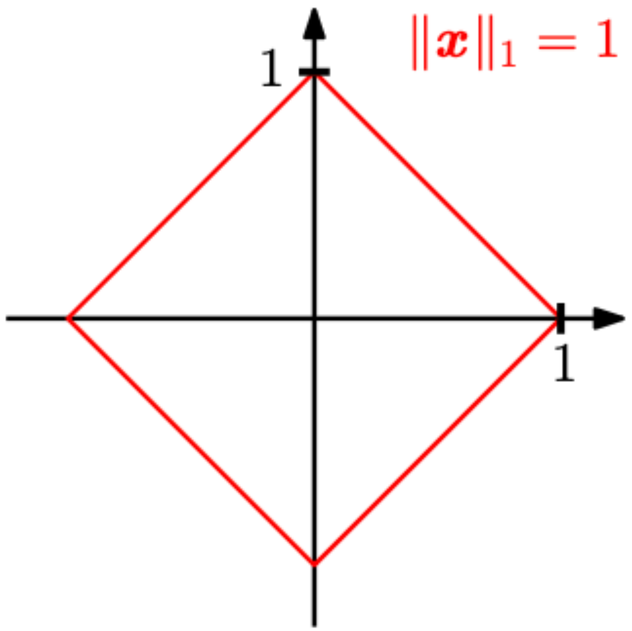
Hence, ℓ_1 is more robust to outliers.

We will learn about probabilistic modelling (taking into account statistics of data) for machine learning in future lectures.

Geometric Intuition of ℓ_1 and ℓ_2 Norm

All vectors $\boldsymbol{x} \in \mathbb{R}^2$ with

same ℓ_1-norm



same ℓ_2-norm

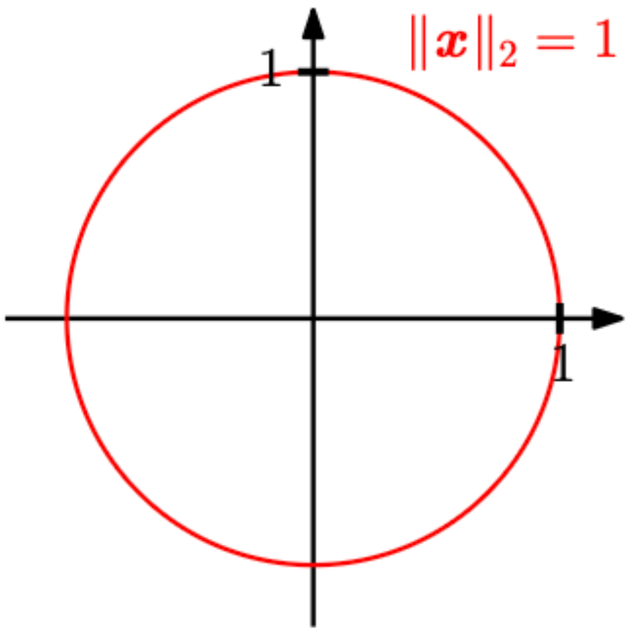


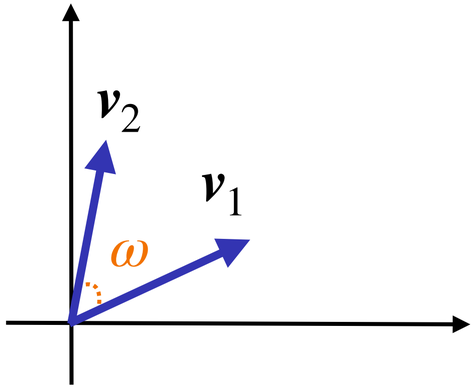
figure from Machine Learning A Bayesian and Optimization Perspective textbook.

Note: Length, Distance, and Angle Can be Induced by Inner Products

ℓ_2-norm $$\|\boldsymbol{x}\|_2 = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$$

Euclidean distance $$d(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_2 = \sqrt{\langle \boldsymbol{x} - \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y} \rangle}$$

Angle ω between vectors $$\cos(\omega) = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle \langle \boldsymbol{y}, \boldsymbol{y} \rangle}}$$



Orthogonal Vectors and Matrix

Orthogonality

Two vectors $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *are* ***orthogonal*** *(*$\boldsymbol{x} \perp \boldsymbol{y}$*) if and only if*

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$$

If additionally $\|\boldsymbol{x}\|_2 = 1 = \|\boldsymbol{y}\|_2$*, then they are* ***orthonormal****.*

Orthogonal Matrix

A square matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ *is an orthogonal matrix if an only if its columns are orthonormal so that*

$$\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{I} = \boldsymbol{A}^\top \boldsymbol{A} \qquad \Rightarrow \qquad \boldsymbol{A}^{-1} = \boldsymbol{A}^\top$$

Symmetric Positive Definite Matrix

A **symmetric** matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is **Positive Definite** (PD) if it satisfies

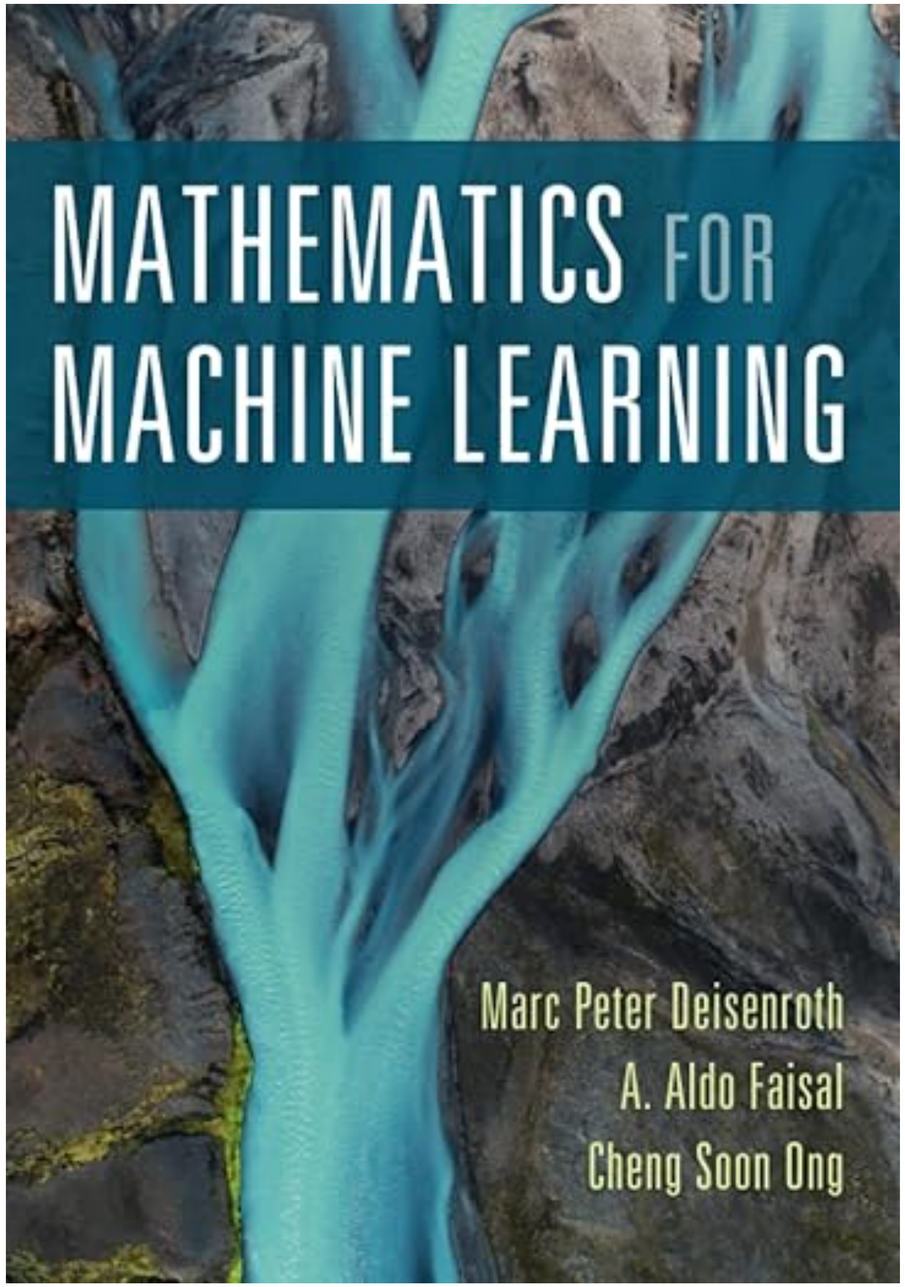
$$\forall \text{ non-zero } \boldsymbol{x} \in \mathbb{R}^n : \quad \boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x} > 0$$

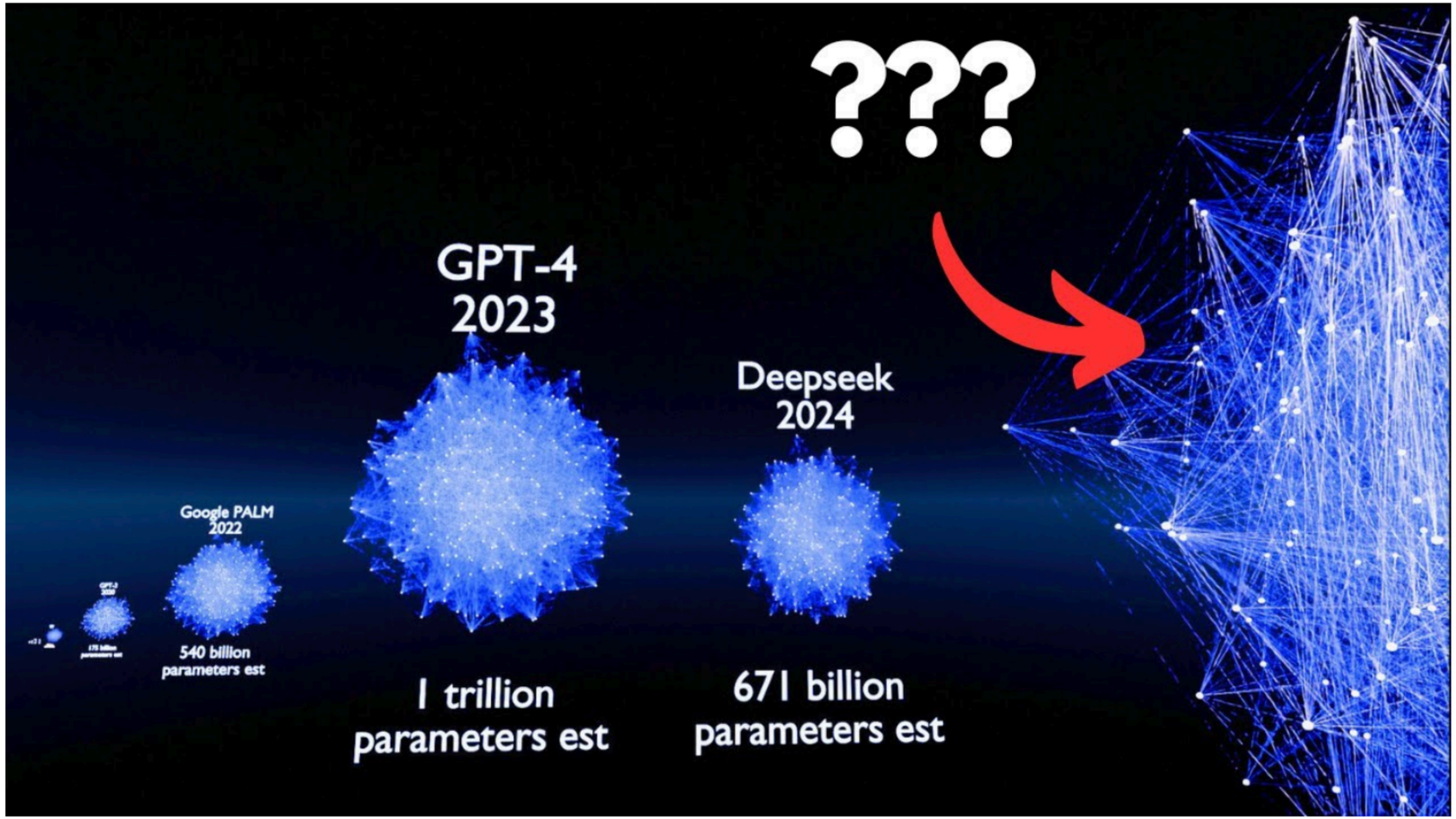
If only $\geq$ holds, then the matrix is called *symmetric positive semidefinite (PSD).*

$\boldsymbol{A} \succ 0$	$\boldsymbol{A} \succeq 0$
PD	PSD

Reference

Chapter 3:
Analytic Geometry





https://www.youtube.com/watch?v=coG8ENwEuz0